

## Part 1: Descriptive Statistics

### 1. DATA QUALITY CHECKS, CLEANING AND DATA ENRICHMENT

#### 1.1 Data quality issues and cleaning steps

Total Number of Rows and Columns: 26000 and 15, respectively.

| Issues                   | Findings                                                                                                                                                                                                                                                                                                                                                                          | Cleaning/ Justifications                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Missing Values           | None                                                                                                                                                                                                                                                                                                                                                                              | No imputation or row deletion was required                                                                                                                                                                                                                                                                             |
| Duplicate transactions   | None                                                                                                                                                                                                                                                                                                                                                                              | No deduplication step required                                                                                                                                                                                                                                                                                         |
| Data types and structure | <ul style="list-style-type: none"><li>– Date Column, <i>tx_date</i>, was stored as an object (string)</li><li>– <i>Store id</i> was stored as a numeric quantity</li></ul>                                                                                                                                                                                                        | <ul style="list-style-type: none"><li>– converted to a categorical variable for date-based analysis (e.g., weekday/weekend effects).</li><li>– converted to a categorical variable because it is an identifier that supports grouped summary statistics and modeling.</li></ul>                                        |
| Data inconsistencies     | None                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                        |
| Outliers                 | <ul style="list-style-type: none"><li>– <i>tx_revenue</i> has <b>1044 outliers</b> - first quartile (Q1) is 4.75, and the third quartile (Q3) is 10.75, giving an <b>IQR of 6.00</b>, and <i>tx_cogs</i> has <b>659 high-value</b> transactions considered as outliers- first quartile (Q1) is 1.14, and the third quartile (Q3) is 3.90, giving an <b>IQR of 2.76</b>.</li></ul> | <ul style="list-style-type: none"><li>– Used the <b>IQR rule</b>(Detects extreme values beyond the interquartile range limits) to flag outliers</li><li>– Not treated as they reflect authentic large transactions - particularly infrequent pastry-box purchases and do not represent data-entry anomalies.</li></ul> |

#### 1.2 Data Enrichment: Definitions and Rationale for Engineered Variables

These engineered variables were selected because they directly support the project’s core objective of evaluating whether the meal deal changes customer behaviour and improves revenue performance. After completing all data cleaning and enrichment steps, including the creation of ‘gross\_margin\_dollars’, ‘gross\_margin\_pct’, ‘has\_food’, ‘has\_beverages’, ‘avg\_item\_price’, ‘total\_food\_count’, and ‘total\_Beverage\_count’—the final analytical dataset contains 26,000 rows and 22 columns.

| Column               | Definition                                                                                     | Rationale                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| avg_item_price       | Transaction revenue divided by items purchased: ( <i>tx_revenue</i> / <i>items_purchased</i> ) | It helps compare per-item <b>spending behaviour</b> between Store0004and Store0014 and assess whether the meal deal affects spending patterns. |
| total_food_count     | Total number of food items in a transaction                                                    | It indicates whether the meal deal increases food purchases and supports <b>food-related behavioural analysis</b> for Store0004and Store0014 . |
| total_beverage_count | Total number of beverages in a transaction.                                                    | It is useful for analyzing <b>beverage-heavy purchases</b> and understanding drink patterns that influence revenue.                            |

### 2. SUMMARY STATISTICS AND VISUALIZATION

#### Transaction revenue (tx\_revenue)

| Statistics           | Combined | Store 04 | Store0014 |
|----------------------|----------|----------|-----------|
| Mean                 | 9.03     | 8.65     | 9.39      |
| Median               | 8.00     | 6.75     | 8.00      |
| Standard Deviation   | 6.20     | 6.08     | 6.29      |
| Skewness Coefficient | 3.30     | 3.18     | 3.41      |

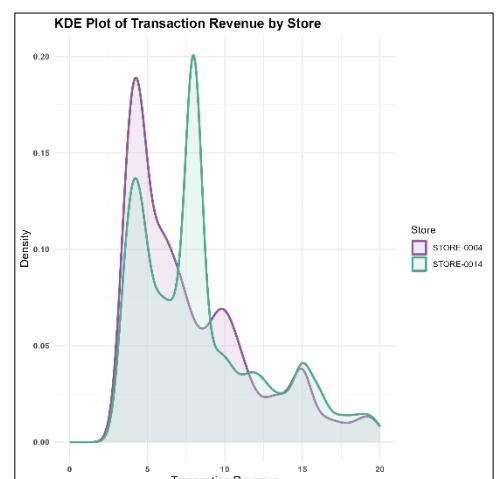


#### BOX PLOT(on left):

Store-0014 reports higher transaction revenue (Median = \$8, IQR = \$5.2–\$11.5) than Store-0004 (Median = \$6.8, IQR = \$4.5–\$10.5) which suggests that Store-0014 consistently earns more per visit.

#### KDE PLOT(on right):

Store-0014 has stronger per-transaction spending as it has higher density in the \$7–\$9 revenue range compared to Store-0004.

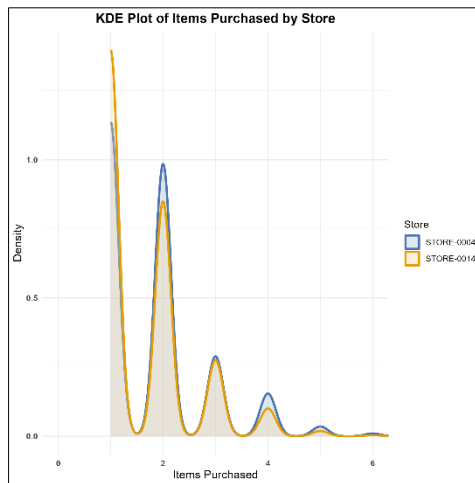


### Items purchased (items\_purchase)

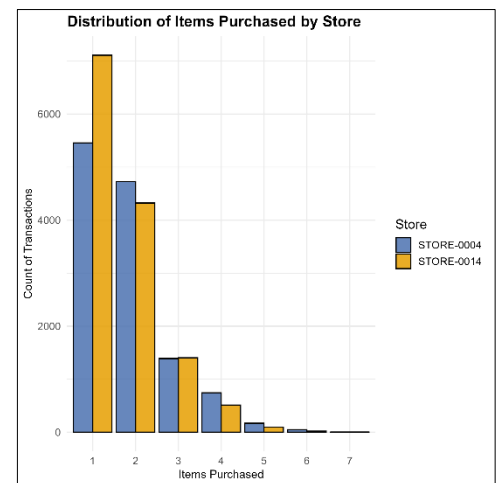
| Statistics           | Combined | Store 04 | Store0014 |
|----------------------|----------|----------|-----------|
| Mean                 | 1.76     | 1.85     | 1.68      |
| Median               | 2.00     | 2.00     | 1         |
| Standard Deviation   | 0.93     | 0.97     | 0.88      |
| Skewness Coefficient | 1.36     | 1.30     | 1.40      |



**BOX PLOT:** Both stores have similar item-purchase distributions with IQR = 1–2, but Store-0004 shows a slightly higher median (2 vs. 1), which suggests marginally larger basket sizes.



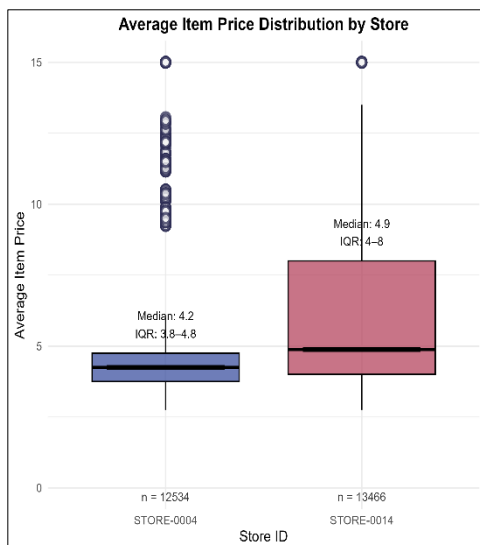
**KDE PLOT:** The KDE distribution for both stores is sharply peaked at 1 item, but Store-0004 shows higher density at 2–3 items, indicating that customers there are more likely to add an extra item.



**BAR CHART** (for better visualisation): Store-0014 has more 1-item purchases, while Store-0004 shows higher counts of 2- and 3-item transactions, supporting the idea that Store-0004 customers tend to buy slightly more items per visit

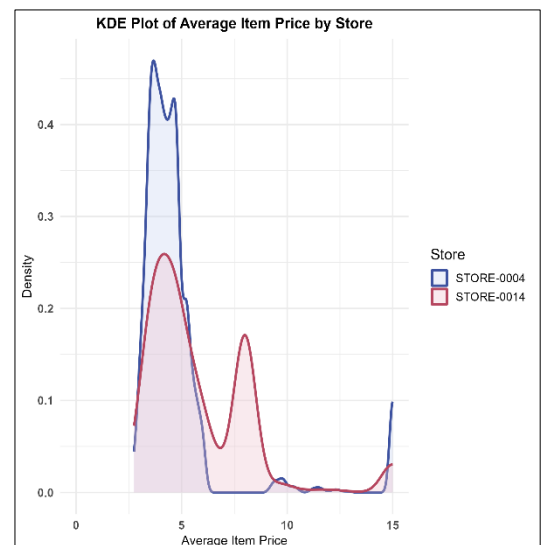
### Average Item Price (avg\_item\_price)

| Statistics           | Combined | Store 04 | Store0014 |
|----------------------|----------|----------|-----------|
| Mean                 | 5.32     | 4.78     | 5.83      |
| Median               | 4.50     | 4.25     | 4.88      |
| Standard Deviation   | 2.53     | 2.32     | 2.61      |
| Skewness Coefficient | 2.38     | 3.47     | 1.80      |



**BOX PLOT**(on left): Customers at Store-0014 tend to buy more expensive items like Pastry Box than at Store-0004, as a reason it has higher average item prices (Median = 4.9, IQR = 4–8) compared to Store-0004 (Median = 4.2, IQR = 3.8–4.8).

**KDE PLOT**(on right): The density curve shifts rightward of Store-0014 with stronger peaks around \$5–\$8, while Store-0004 is concentrated around \$4–\$5, reinforcing that Store-0014 sells higher-priced items more frequently.



## 3. CONDITIONAL PROBABILITY

### • P(purchases 3+ items)

Store-0004 customers purchase 3+items more often, which indicates “basket-building” behaviour is stronger than Store-0014. This suggests Store-0004 customers are adding extra items even without a promotion. From a business point of view, Store-0004 is a good place to encourage “add one more item” as as people already tend to buy more.

| Store ID   | P(purchases 3+ items) |
|------------|-----------------------|
| STORE-0004 | 0.187                 |
| STORE-0014 | 0.151                 |

• **P(purchases food | purchases beverage)**

Store-0014 customers are more likely to buy a beverage with food than Store-0004. The meal deal appears to successfully nudge customers toward combined food-and-drink purchases, exactly what the promotion aims to achieve.

| Store ID   | P(food   beverage) |
|------------|--------------------|
| STORE-0004 | 0.493              |
| STORE-0014 | 0.551              |

• **P(purchases pastry box | weekend) and P(purchases pastry box | weekday)**

Pastry box purchases nearly double on weekends at both stores, regardless of treatment. This shows that customers view pastries as a weekend treat or something to share. Stores can leverage this behaviour by offering ‘weekend-only pastry specials’ or enhancing bakery displays to drive higher weekend sales.

| Store ID   | P_pastry_weekend | P_pastry_weekday |
|------------|------------------|------------------|
| STORE-0004 | 0.0795           | 0.0452           |
| STORE-0014 | 0.0852           | 0.0464           |

Part 2: Inferential Statistics

4. HYPOTHESIS TESTING

To evaluate whether the meal-deal promotion increased profitability, the two stores were evaluated on the metric “Average Daily Gross Profit.” The hypotheses were defined as follows:

- $H_0: \mu_{14} = \mu_4$ . There is no difference in average daily gross profit between Store0014 and Store 04.
- $H_a: \mu_{14} > \mu_4$ . Store0014 has a higher average daily gross profit than Store 04.

Using 28 daily profit observations per store, a two-sample t-test (unequal variances) showed that Store0014 reported a higher mean daily gross profit of \$2,921.36 than Store 04, which reported \$2,685.78, a difference of approximately \$236 per day. The one-tailed p-value was 0.1442, which is greater than the 0.05 significance level. Therefore, **I fail to reject the null hypothesis, indicating that the difference is not statistically significant.** However, the statistical test does not confirm a significant difference; **the profit gap still matters in real business terms.** Store0014 makes about \$236 more profit per day, which adds up to over \$6,600 in just 28 days. This is a meaningful increase for daily operations. These results suggest the meal-deal promotion could be helping, but we would need more time or more store data to be fully confident.

5. REGRESSION ANALYSIS

A multiple linear regression model was built to predict **average daily revenue** using both categorical and continuous store characteristics. The final selected model includes:

- **Categorical variables:** group (Control/Treatment), location\_type (Urban, Suburban, Highway), has\_drive\_thru (Yes/No)
- **Continuous variables:** store\_size\_sqft, years\_operating, local\_population\_k

**Parking spaces** showed a *very small and insignificant t-value* ( $t = -0.43$ ) as well as an *insignificant p-value* ( $0.671 > 0.05$ ), making it the least influential predictor and the only variable removed from the final model.

Model

$$avg\_daily\_revenue \sim group + location\_type + has\_drive\_thru + store\_size\_sqft + years\_operating + local\_population\_k$$

COEFFICIENT TABLE

| Predictor Variable    | Type        | Estimate  | Std. Error | t value  | Pr(> t ) | Significance |
|-----------------------|-------------|-----------|------------|----------|----------|--------------|
| (Intercept)           | (Baseline)  | 2148.0455 | 224.56155  | 9.565509 | < 2e-16  | ***          |
| groupTreatment        | Categorical | 420.95898 | 76.4335515 | 5.507516 | 1.16E-07 | ***          |
| location_typeSuburban | Categorical | -645.9435 | 92.0508877 | -7.01724 | 3.78E-11 | ***          |
| location_typeHighway  | Categorical | -1001.773 | 106.869445 | -9.3738  | < 2e-16  | ***          |
| has_drive_thruNo      | Categorical | -696.3169 | 81.7043212 | -8.5224  | 4.55E-15 | ***          |
| store_size_sqft       | Continuous  | 0.6844739 | 0.09923325 | 6.897626 | 7.45E-11 | ***          |
| years_operating       | Continuous  | 34.320975 | 9.30943013 | 3.686689 | 0.000296 | ***          |
| local_population_k    | Continuous  | 10.590605 | 1.78947503 | 5.918275 | 1.47E-08 | ***          |

*Note: The intercept reflects the expected daily revenue for a baseline store (Urban location, Control group, Has drive-through, with size, population, and years set to zero).*

Model Fit:

- $R^2 = 0.577$ , Adjusted  $R^2 = 0.561$
- F-statistic = 37.37,  $p < 0.001$
- Residual Standard Error  $\approx 525$

Key Findings and Business Interpretation

Stores with the meal-deal promotion earned **about \$420 more per day**, and this effect is **statistically strong** high t-value ( $t = 5.51$ ,  $p < 0.001$ ) and unlikely due to chance. The model explains approximately **58% of the variation** in daily revenue and the adjusted  $R^2$  of 56% suggests the model fits well without overfitting ( $R^2 = 0.577$ ; adjusted  $R^2 = 0.561$ ), indicating a good fit. This means the treatment effect remains meaningful even after accounting for key store characteristics. The regression shows that several store characteristics strongly influence average daily revenue. **Location type is the most influential predictor** as it has the *largest magnitude t-value* ( $t =$

–9.37) in the model, with highway stores earning about **\$1,003 less** per day, and suburban stores earning \$646 less ( $t = -7.02$ ). Stores without drive-throughs also earn significantly less, i.e., –\$696/day ( $t = -8.52$ ), confirming that convenience strongly drives sales. Larger stores perform better, with each additional square foot adding \$0.68 to revenue ( $t = 6.90$ ). Stores that have been operating longer earn about \$34 more per year ( $t = 3.69$ ), and denser local populations add \$10.59 per 1,000 residents ( $t = 5.92$ ).

### Business Implications

These findings highlight clear strategic priorities:

- Since urban, high-traffic, and drive-through-equipped stores naturally generate higher revenue, the company should continue investing in these locations and consider expanding store size where feasible.
- The high  $t$ -value for the meal-deal promotion demonstrates that it provides meaningful revenue uplift independent of store type, supporting chain-wide expansion.
- By scaling the promotion first in urban, high-population, and drive-through stores, the company can capture the highest immediate uplift, while still expecting positive results in other settings.
- Overall, the regression provides compelling evidence that the meal deal adds incremental value and should be implemented broadly, with targeted emphasis on stores that already have strong structural advantages.

## 6. SYNTHESIS AND RECOMMENDATIONS

### Insights About Customer Purchasing Patterns from Transaction-Level

- Store0014 customers spend more per transaction.**  
The median revenue of Store-0014 is \$8.05 vs. \$6.80 at Store0004, and its average item price is \$5.80 vs. \$4.80, which clearly shows average spending is higher in the meal deal store. Store0014 customers gravitate toward higher-priced items like pastry boxes more frequently, as the count of pastry purchased is 733 at Store0004 and 826 at Store0014, leading to more transactions in the \$7–\$9 range.
- Store0004 customers naturally buy more items.**  
There is a greater probability of 3+ item purchases (18.7% vs. 15.1%) at Store0004, as it shows a higher median basket size (2 items vs. 1 item at Store 0014). Store0004 has noticeably more transactions in the 2–3 item range. This pattern highlights stronger organic multi-item purchasing behaviour even without the meal-deal promotion. However, meal deals count as one purchase (not as separate food and beverage items), so customers may be stopping at the meal deal instead of adding more items to their basket.
- Food–beverage bundling is stronger at Store0014.**  
Store 0014 shows a higher probability of purchasing food when a beverage is bought (**55.1% vs. 49.3%**), indicating stronger bundling behaviour encouraged by the meal-deal environment. This contributes to higher transaction revenue because paired purchases typically result in larger-dollar transactions.
- Pastry boxes act as weekend-driven purchases across both stores.**  
Pastry box purchases nearly double on weekends regardless of the meal deal—Store 04 sees 7.9% weekend purchases vs. 4.5% on weekdays, while Store 14 shows 8.5% vs. 4.6%. This consistent pattern across both stores tells us that weekend pastry demand is strong and independent of promotions, representing an opportunity to boost sales through separate weekend marketing.

### Meal Deal Promotion and Store Performance Driver Recommendations

- Develop “Weekend-Only” Pastry Bundles**  
On weekends, pastry box purchases shoot up to double that of weekdays, which shows consistency in time-based demand spike. In this scenario, introducing limited-time weekend pastry bundles like Pastry box + 2 Drinks will help monetize this predictable surge. Inventory planning and staffing should be adjusted according to the weekend to maximize profitability.
- Redesign the Meal Deal as a Tiered Structure to Prevent Basket Capping**  
Store 04 has a higher share of 3+ item purchases (18.7% vs. 15.1% at Store 14). Because the meal deal counts as just one unit instead of separate food and beverage items, it may unintentionally limit how many items customers add. A tiered structure like “\$1 off for food + beverage, \$2 off when adding a third item” helps keep the bundle attractive while still encouraging larger baskets.
- Prioritize Urban Locations with Drive-Through Infrastructure**  
The regression shows that urban stores and drive-through locations earn much more revenue each day. The combined effect means an urban store with a drive-through generates ~\$1,700 more daily revenue than a highway store without a drive-through, \$620,500 annual difference. This structural advantage is independent of promotional activity. Hence, rolling out the meal deal and focusing marketing efforts in these locations will give the highest payoff.
- Leverage Store Tenure Through Loyalty and Community Programs**  
Every extra year a store has been open adds about \$34 more in revenue per day, which adds up to roughly \$125,000 a year for a 10-year-old store. This shows that long-time customers and daily routines really matter. Instead of shutting down older stores that aren’t performing well, it’s better to invest in them with small renovations, updated layouts, or loyalty programs to make the most of the customer base they’ve built over the years.

## CONCLUSION

Overall, the meal deal should be expanded chain-wide, but in phases, not in all stores at once. Store 14 consistently earns more; its median revenue is \$8.05 vs. \$6.80 at Store 4, and its average item price is almost a dollar higher, which is clearly visible in the boxplots and KDE graphs. While the two-store hypothesis test wasn’t statistically significant, the larger 200-store regression shows a strong and reliable treatment effect of about \$420 extra revenue per day, even after adjusting for location, drive-through status, size, and population. Combined with higher food-beverage pairing rates (55.1% vs. 49.3%) and weekend pastry spikes across both stores, the evidence strongly supports expanding the meal deal, especially in urban and drive-through locations where the uplift will be highest.